What do Venture Capitalists Know?

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Abstract

This paper aims to examine to what extent venture capitalists are informed about the evolution of start-up value and how predictive ability differ across VCs. The main empirical challenge is the lack of data. Researchers do not observe all the data available at the time of decision-making, nor VC expectations that directly determines decision. We adopt a moment inequality approach to set identify model parameters. We found that predictive accuracy increases with experience and network depth, indicating a potential positive feedback loop when VCs jointly invest.

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1 Introduction

Startups, largely funded by venture capital (VC hereafter), are a critical source of innovation. For example, 57% of new drugs approved in the United States in 2023 originated at startups (Economist 2024). Consequently, VC investment decisions play a key role in driving economic growth and technological progress (Kortum and Lerner 2000, Samila and Sorensen 2011). However, information asymmetries between VCs and startups regarding firm quality may lead to the underfunding of ventures with high potential. For example, Bessemer Venture Partners, one of the largest VC firms, declined early investment opportunities in companies such as Airbnb, Apple, eBay, Facebook, Tesla, and Zoom, citing high uncertainty. To inform policies aimed at promoting VC investment, it is essential to understand how VCs make investment decisions under uncertainty.

This paper investigates the extent to which VCs are informed about the evolution of startup value and how predictive abilities vary across VC investors. A central empirical challenge is that investment decisions depend on VCs' expectations, which are not directly observable. Researchers must therefore impose assumptions about how these expectations are formed. One common assumption is perfect foresight, in which agents' expectations align with future realizations. However, Bessemer's missed opportunities suggest that this assumption may be unrealistic. Similar concerns have been raised in other areas of economics, where evidence shows that agents form expectations imperfectly—for example, by over- or underestimating the returns to education (Cunha and Heckman 2007). Rather than developing identification strategies that are robust to alternative assumptions (Haile and Tamer 2003), we build a behavioral model to construct odds-based and revealed preference conditions. This allows us to estimate VCs' decision rules while specifying only a subset of their information set, following Dickstein and Morales (2018) and Bombardini et al. (2023).

We model venture capital (VC) financing as a take-it-or-leave-it offer from the startup to the investor. At the time of the investment decision, VCs do not observe the future value of the startup and must rely on their expectations. In our framework, all VCs share a common observable information set but also receive private signals drawn from heterogeneous distributions, which vary with investor characteristics. We then estimate how the precision of these private signals correlates

with VC characteristics. The model assumes no strategic interaction among VCs, and we provide suggestive evidence supporting the plausibility of this assumption. More effort will be invested in relaxing this assumption in the future.

Our empirical analysis draws on data from PitchBook, one of the most comprehensive sources on VC activity (Retterath and Braun, 2020). The dataset includes detailed information on funding rounds, equity stakes, and CEO characteristics, spanning from seed financing to exit (via IPO, M&A, LBO, or failure). We supplement this dataset with Compustat stock prices and HUPD patent filings. Although financial metrics are missing for many deals, our estimation approach remains robust due to its structural flexibility. Additionally, we use machine learning classification methods to identify VCs who considered but ultimately declined to invest in specific deals.

We begin with reduced-form evidence, constructing performance metrics from VC investment histories and examining how they correlate with investor characteristics. Given that this evidence is noisy and inconclusive, we proceed with a structural estimation of the model using the approach proposed by Dickstein and Morales (2018). We flexibly estimate confidence intervals for the parameters using a simulated moment inequality framework. The results indicate that both experience and network breadth improve the predictive accuracy of VCs. These findings suggest that policies promoting venture investment may create positive feedback loops and generate long-term benefits by enhancing the information efficiency of the VC sector.

Related Literature This study contributes to several strands of literature. First, it relates to the longstanding discussion on venture capital (VC) performance. A large body of work has documented the persistence of VC performance across time. For example, Kaplan and Schoar (2005) use data from the 1980s and 1990s, and Hochberg et al. (2014) extend the sample through 2002, both finding evidence of persistent performance. More recently, Harris et al. (2023) show that VC performance is more persistent than that of buyout funds. Scholars have also explored the underlying mechanisms behind this persistence. Hochberg et al. (2007) demonstrate that better-networked VCs perform significantly better, while Nahata (2008) develop a measure of VC reputation and show that startups backed by more reputable VCs tend to outperform. Gompers et al. (2009) highlight the role of specialization and experience in driving VC success. This paper contributes to

this literature by examining whether commonly studied VC characteristics—such as experience, network centrality, and specialization—are associated with informational advantages, and whether such advantages contribute to performance persistence.

Second, this study contributes to the literature on the determinants of VC investment decisions. Da Rin et al. (2013) provide an early survey of this topic. Using a randomized field experiment, Bernstein et al. (2017) show that information about founding teams plays a pivotal role in attracting VC interest. Similarly, Gompers et al. (2020) use survey data to show that VCs place the greatest weight on the management team during the screening process and attribute investment outcomes more to the team than to the product or business model. While these studies rely on detailed data collection to infer VC beliefs, this paper adopts the methodology developed by Dickstein and Morales (2018), which allows researchers to infer agents' information sets without directly observing them.

Third, this paper contributes to the growing literature that incorporates agent forecasts into empirical economic analysis (e.g., Allen and Arkolakis 2014; Ameriks et al. 2015; Dickstein and Morales 2018; Bombardini et al. 2023). Dickstein and Morales (2018) examine how exporters' expectations influence their entry decisions, finding that larger firms tend to be better informed. Bombardini et al. (2023) apply a similar framework to assess what information U.S. legislators possessed when voting on policies related to the China shock. This study is the first to apply this class of methods to the venture capital setting, enabling a novel analysis of how investor characteristics relate to private signal precision and decision-making under uncertainty.

2 Model

We model venture capitalists' (VCs') investment decisions, where the unit of observation is a deal in which a VC satisfies a startup's funding requirement in exchange for equity. The market consists of two participant groups, VCs and startups, linked via an exogenously determined bipartite graph. Each startup must complete an unknown number of funding rounds, n_j, before its true value, ξ_j, is revealed through an exit event such as an IPO, LBO, or failure.

Each funding round t for startup j is characterized by a tuple VC_{jt}, k_{jt}, p_{jt}, where VC_{jt} denotes

the set of potential VC investors selected exogenously among those linked to startup j, k_{jt} is the capital required, and p_{jt} is the 'price per share'. The offer $\{k_{jt}, p_{jt}\}$ is take-it-or-leave-it: a VC may purchase $a_{jt} = k_{jt}/p_{jt}$ shares by investing k_{jt}. We assume entrepreneurs play a passive role, sequentially presenting this offer to potential VCs in VC_{jt} in random order. Once a VC accepts the offer, the round concludes and the startup advances to round $t + 1$.

VCs are assumed to be myopic, evaluating investments based solely on round-by-round returns. If VC i accepts the offer in round t for startup j, they receive a capital gain of $\pi_{ijt} = a_{jt}(\delta p_{j,t+1} - p_{jt})$; otherwise, they receive no payoff. This assumption reflects common industry practices: (i) VCs often partially exit by selling equity in later rounds; (ii) many VCs specialize in specific investment stages rather than the final outcome; and (iii) startup valuations, particularly in sectors like pharmaceuticals, can exhibit large discrete jumps.

However, at the time of investment, the future price $p_{j,t+1}$ is unknown, so VCs must form expectations. Let J_{ijt} denote the information set available to VC i at round t for startup j, and let $\nu_{ijt} \sim \mathcal{N}(0, \sigma_i^2)$ denote a private signal received by the VC. The expected return conditional on information and the signal is:

$$E(\pi_{ijt}|J_{ijt}, \nu_{ijt}) = a_{ij(}\delta E(p_{j,t+1}|J_{ijt}) + \nu_{ijt} - p_{ijt}).$$

Additionally, assume the signal precision depends on VC characteristics X_i via:

$$\frac{1}{\sigma_i^2} = exp(X_i'\beta).$$

Thus, a VC's expectation comprises a systematic component, $E(p_{j,t+1}|J_{ijt})$, and an idiosyncratic component, ν_{ijt}, whose variance depends on X_i. The econometrician's goal is to estimate β, which captures how VC characteristics influence predictive ability regarding startup success. We acknowledge that this model assumes strategic interactions away, suggestive evidence and discussions of such assumption is postponed to the next section.

Define the investment decision as $d_{ijt} = \mathbf{1}E(\pi_{ijt}|J_{ijt}, \nu_{ijt}) \geq rp_{jt}$, where r is the required

return in the VC industry. Then:

$$d_{ijt} = \mathbf{1}\{\delta E(p_{j,t+1}|J_{ijt}) + \nu_{ijt} - (1+r)p_{ijt}) \geq 0\}.$$

The probability that VC i invests in startup j at round t, conditional on information set J_{ijt}, is

$$\begin{aligned} P(d_{ijt} = 1|J_{ijt}) &= \int_{\nu} \mathbf{1}\{\delta E(p_{j,t+1}|J_{ijt}) - (1+r)p_{ijt} + \nu) \geq 0\}(\frac{1}{\sigma_i})\phi(\frac{\nu}{\sigma_i})d\nu \\ &= \Phi(exp(\frac{1}{2}X_i'\beta)(\delta E(p_{j,t+1}|J_{ijt}) - (1+r)p_{ijt})), \end{aligned} \tag{1}$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the standard normal probability density and cumulative distribution functions, respectively.

3 Data

3.1 Pitchbook

PitchBook is a leading data provider for information on private companies, particularly those backed by venture capital (VC)[1]. It offers detailed information on VC deals and the startups involved. On the VC side, we collect deal-level information including deal size, equity share acquired by the investor, deal date, deal status, and financing stage. From the startup side, we gather data on founding year, location, industry, and other firm characteristics. VC characteristics—such as experience, network centrality, reputation, and specialization—are also constructed using PitchBook data.

We begin with the full universe of VC deals in North America recorded in PitchBook, totaling more than 390,000 deals from the 1970s through 2024. We restrict the sample to deals involving U.S.-based startups after January 1, 2000. To compute price-related variables, we divide the deal size (i.e., the dollar amount invested in a given round) by the percentage of equity acquired by the investor, thereby recovering an implied valuation. We retain deals for which both deal size and

[1]E.g. Retterath and Braun (2020); Gompers et al. (2021); Ewens et al. (2022)

ownership percentage are available. We further exclude startups that appear in PitchBook as having only one recorded financing round, ensuring that we have sufficient longitudinal information for analysis. These restrictions yield a final sample of 40,591 deals.

Next, we collect exit information for the remaining startups. Specifically, we examine each firm's current business status, which can be: publicly held, merged/acquired, privately held, or out of business. For publicly held firms—those that completed an IPO—we collect IPO-related data to calculate exit valuations. We obtain the first-day closing market capitalization from CRSP, Compustat, and Datastream, which together cover the vast majority of VC-backed IPOs. For remaining IPOs not covered by these sources, we hand-collect data whenever possible. For mergers and acquisitions (M&A), we use exit valuations from PitchBook's buyer information file, which provides details of M&A transactions. For startups marked as out of business, we assume their exit valuation drops to zero.

Table 1 presents summary statistics for the deals in our sample. On average, a VC deal involves an investment of \$40.77 million in exchange for a 37.84% equity stake in the startup, yielding an average round-wise return exceeding 170%. However, the distribution of deal characteristics is highly skewed, as evidenced by the large gap between the mean and the median. The median deal involves a significantly smaller investment of \$5.6 million for a 27.45% equity stake, with a round-wise return just under 80%.

VARIABLES	N	mean	p25	p50	p70
$percentacquired_{j,t}$	48,102	37.84	16.96	27.45	45.24
$dealsize_{j,t}$	45,668	40.77	1.999	5.558	15
$p_{j,t+1}$	30,615	1.483	0.133	0.350	0.920
$p_{j,t}$	45,668	1.073	0.0837	0.223	0.620
$r_{j,t-1\to t}$	30,577	1.729	0.0831	0.792	1.913
$r_{j,t\to t+1}$	30,570	1.733	0.0837	0.794	1.917
$p_t^{round,year}$	48,102	1.173	0.274	0.581	1.385
$p_t^{industry,year}$	48,028	1.151	0.501	0.752	1.243
$patent_{j,t,cum}$	47,942	3.413	0	0	2

This table shows summary statistics for the sample of VC deals.

Table 1: Summary Statistics

3.2 Infer potential investors

One limitation of PitchBook is that it reports only the VCs that ultimately invested in a deal, but not those who were potentially involved but chose not to invest. In other words, we observe only $d = 1$ decisions, while $d = 0$ decisions are unobserved. In practice, VCs screen a large number of startups and invest in only a small fraction, with an investment rate of roughly 1%. To estimate a discrete choice model, we therefore need to impute potential investor-startup pairs that reflect such unobserved $d = 0$ decisions. To this end, we adopt a graph neural network (GNN) approach.

GNNs are neural networks designed to operate on graph-structured data. They learn node representations by iteratively aggregating information from neighboring nodes and their features. Intuitively, the GNN clusters VCs and startups into groups with similar—potentially unobserved—characteristics based on historical investment behavior. If we observe that many VCs from one group frequently invest in startups from another group, GNN infers that other VCs in the same group are likely potential investors in startups in the other group, even if such investments are not observed in the data.

Formally, a graph $G = (N, E)$ consists of nodes N (VCs and startups) and edges E (observed investments). Consider a startup node s. The set of VC nodes historically connected to s via observed investments is denoted N_s. Each node is associated with a feature vector, characteristics of the startup or VC, which is aggregated and updated across layers of the GNN. We adopt a convolutional architecture, where the feature update for node s at layer l is given by

$$h_s^l = max\{0, \sum_{n \in N_s, s} \frac{1}{\#(N_s + 1)} W^l h_s^{l-1}\}$$

where W^l is a weight matrix trained to minimize the discrepancy between predicted and observed edges. After several layers of aggregation, the final node embeddings are used to compute edge scores—interpreted as the likelihood of a VC-startup match.

In implementation, the node features include location, industry sector, and investor/startup size (measured by total capital invested or raised). We further restrict the pool of potential investors to VCs who led at least one deal in the same year. To manage computational burden, we randomly

sample a set of negative edges (unobserved no investment VC-startup pairs) from among the high-scoring predictions of the GNN, proportional to the numbers of edges linking startups. These sampled pairs are appended to the observed dataset as $d = 0$ observations for estimation.

4 Reduced-form Evidence

4.1 Performance and characteristics

In this section, we present reduced-form evidence on the relationship between VC characteristics and performance. We begin by defining three performance metrics, *regret*, *rank*, and *excess*, based on the imputed VC-startup choice sets. *Regret* is the ex post annualized log return difference between the best-performing startup in a VC's choice set and the one the VC selected. It captures the potential upside the VC missed. *Rank* is the percentile ranking of the chosen startup's return within its choice set. *Excess* is the log return difference between the chosen startup and the average return of the VC's choice set.

Together, these three metrics offer a comprehensive assessment of VC performance, particularly given the high-variance nature of venture investments. Consider two hypothetical 5-startup choice sets with return profiles $\{1, 1, 2, 5, 10\}$ and $\{1.7, 1.8, 1.9, 2, 3\}$, where both VC 1 and VC 2 select the second-best startup. While regret suggests that VC2 (-1) underperforms VC1 (-5), rank indicates they are equally skilled (both at the 80th percentile), and excess reveals that VC1 (1.2) outperforms VC2 (-0.08). This example illustrates why evaluating all three metrics provides a more nuanced understanding of VC decision quality.

We estimate the following reduced-form equations to examine how VC performance correlates with observable characteristics:

$$\begin{aligned} measure_{it} &= \beta X_{it} + \epsilon_{it}, \\ measure_{it} &= \beta X_{it} + \alpha_i + e_{it}, \end{aligned} \tag{2}$$

where i indexes VC and t indexes months. The first equation captures cross-sectional correlations between VC characteristics and performance. The second includes VC fixed effects to isolate the

	Mean	Std	Min	Max
Performance				
regret	0.56	0.92	0	6.44
rank	0.49	0.31	0	1
excess	-0.13	0.87	-2.90	3.66
VC				
CVC	0.10	0.30	0	1
Experience	139.91	216,59	0	2088
Network	0.01	0.01	0	0.11
Size	42.13	30.63	1	135
Startup				
price	1.05	5.87	0.00	575
return	-0.23	0.90	-3	3.56
Count	254.89	114.30	13	575

This table shows summary statistics for the sample of VC performance metrics.

Table 2: Summary Statistics

within-VC effects of changing characteristics over time.

Table 2 summarizes the variables used in these regressions. On average, the best startup in a VC's choice set outperformed the chosen one by 75% ($\bar{regret} = 0.56$). The selected startup typically ranks near the median ($\bar{rank} = 0.49$) and underperforms the average return in the set by about 12% ($\bar{excess} = -0.13$). These averages mask considerable heterogeneity. On the VC side, about 10% are corporate-backed. The average VC has completed 140 deals over 17 years and faces choice sets with an average of 42 startups. Conversely, each startup faces potential investment from approximately 255 VCs.[2]

The regression results are presented in Table 3. We begin with corporate VCs (CVCs). Across all three performance metrics, we find consistent evidence that CVCs underperform traditional VCs. This aligns with the view that corporate venture arms often pursue strategic or exploratory investments rather than maximizing financial return. Regarding experience, both rank and excess improve with more market exposure, suggesting that experienced VCs make better investment decisions. However, regret shows a negative association with experience. This could reflect increased risk aversion among mature VCs, leading them to avoid high-variance opportunities that might of-

[2]Business estimates suggest that VCs invest in approximately 1% of the startups they evaluate.

	(1) regret	(2) regret	(3) rank	(4) rank	(5) excess	(6) excess
CVC	0.047*		-0.023***		-0.070***	
	(0.028)		(0.009)		(0.024)	
Experience	0.001**	0.023***	0.000***	0.007***	0.001**	0.10***
	(0.000)	(0.004)	(0.000)	(0.001)	(0.000)	(0.004)
Network	-0.346	4.169**	1.796***	-0.394	2.154***	-2.354
	(0.785)	(1.641)	(0.248)	(0.516)	(0.694)	(1.469)
Size	0.002***	-0.001**	-0.001***	0.000	0.001***	0.001
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
VC FE	No	Yes	No	Yes	No	Yes
R^2	0.003	0.178	0.012	0.219	0.008	0.174
Observations	13123	11345	13123	11345	13123	11345

Notes: The dependent variable $regret$ is the log return difference between the best startup and invested startup, $rank$ is the percentile rank of the invested startup, $excess$ is the log return difference between the invested startup and the choice set mean. The independent variable CVC is an indicator variable denoting the VC is corporate-backed, $Experience$ is the number of previous invested deals, $Network$ is the number of co-investors divided by all investors, $Size$ is the size of choice set..
Standard errors in parentheses are heteroskedasticity robust. Significance levels: * if $p < 0.10$, ** if $p < 0.05$, *** if $p < 0.01$

Table 3: Performance and VC characteristics

fer higher returns. The role of network is nuanced. Cross-sectional comparisons (Columns 3 and 5) show that broader networks are associated with better performance in terms of rank and excess. However, fixed-effects estimates (Columns 4 and 6) suggest that expanding a network over time is associated with lower, though statistically insignificant, performance. One interpretation is that growing networks may draw VCs into unfamiliar sectors where they co-invest with less precision or domain knowledge.

Overall, these findings motivate a more structural exploration of the mechanisms behind performance differences. In the next section, we examine whether better performance stems from deeper information sets or improved predictive accuracy.

4.2 Winner's curse

Before proceeding to structural estimations, we investigate empirical patterns to assess the validity of our modeling assumptions. Specifically, our model assumes the deal price is exogenously given, and VCs private signal only affects their willingness to invest, but not the price they pay. It is different from a common value auction where VCs determine price considering strategic interactions and the winner's curse. In practice, a VC might infer a startup's quality based on how many other investors have been attracted to the startup and update their valuation accordingly. If there is weak or no evidence for winner's curse in the venture market, the simplifying assumption of exogenous price and no strategic interactions is less of a concern.

To test for the presence of a winner's curse, we estimate the reduced-form equation $y = \beta x$, where y represents either the investment price or realized return, and x denotes a startup's popularity, measured by the number of imputed VCs associated with the startup (**?**). The logic is as follows: if a winner's curse is present, we would expect a positive relationship between price and popularity—startups that attract more interested VCs should receive higher bids, reflecting potential overvaluation.

Table 4 reports the results. Columns (1) and (3) present the cross-sectional evidence, while Columns (2) and (4) include startup fixed effects to control for unobserved startup characteristics. Columns (1) and (2) show statistically significant negative effects between price and VC count. A one standard deviation increase in VC pool size (114) is associated with roughly a 0.2 standard deviation decrease in price.

Another testable implication of the winner's curse is that the return to the winning bid is inversely related to the level of competition (**?**). However, Columns (3) and (4) show the opposite: returns are positively associated with the contemporaneous VC pool size. This pattern is inconsistent with the winner's curse hypothesis.

One important limitation of the preceding analysis concerns the potential endogeneity of the number of VCs associated with a startup. If venture capitalists are drawn to startups based on unobserved characteristics that also influence price or returns, the estimates may be subject to bias. While the inclusion of fixed effects helps account for time-invariant startup characteristics,

	(1) price	(2) price	(3) return	(4) return
Count	-0.012***	-0.016**	0.001***	0.001**
	(0.002)	(0.008)	(0.000)	(0.000)
Count$\hat{2}$	0.000***	0.000**	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Startup FE	No	Yes	No	Yes
R^2	0.009	0.405	0.020	0.432
Observations	12009	6833	12009	6833

Notes: The dependent variable $price$ is 1% value of a startup, $return$ is the annualized log return. The independent variable $Count$ is the number of VCs paired with startup, $Count^2$ is the square of $Count$
Standard errors in parentheses are heteroskedasticity robust. Significance levels: * if $p < 0.10$, ** if $p < 0.05$, *** if $p < 0.01$

Table 4: Performance and VC characteristics

time-varying unobservables may still confound the results. Accordingly, the evidence presented in Table 4 should be interpreted as suggestive rather than definitive with respect to the presence of a winner's curse.

A second caveat relates to measurement error. The size of the VC pool is imputed using graph neural network (GNN) predictions, which may introduce additional noise or bias. As a result, the findings in Table 4 may partially reflect errors inherent in the GNN estimation procedure.

Nonetheless, the suggestive patterns that price is not very responsive to the number of VC paired with provide some reassurance that modeling price as exogenous and excluding strategic interactions among VCs may not distort the core findings too much. In future work, we aim to extend the model to explicitly incorporate the price formation process and account for potential strategic considerations among investors.

5 Structural Approach

In this section, we describe the empirical strategy used to estimate the model parameters. We begin by presenting the canonical maximum likelihood estimator, which relies on the assumption

that the information sets VCs use to form expectations are known and observable to the econometrician. This assumption is clearly strong, especially in settings like ours, where detailed financial metrics are unavailable. To address the potential bias introduced by this assumption, we consider an alternative approach based on moment inequalities, following Dickstein and Morales (2018). This method is more flexible with respect to assumptions about information sets and produces more reliable estimates than the naive MLE in settings with incomplete information.

5.1 Perfect knowledge of information set

When the econometrician's assumed information set J^a_{ijt} matches the VCs' true information set J_{ijt}, the true parameter vector β^* can be estimated using a standard maximum likelihood estimator:

$$\max_{\beta} L(\beta|d, J^a, X) = \sum_{i,j,t} d_{ijt} \ln(P(d_{ijt} = 1|J^a_{ijt}, X_{ijt}; \beta)) + (1 - d_{ijt}) \ln(P(d_{ijt} = 0|J^a_{ijt}, X_{ijt}; \beta)) \tag{3}$$

where $P(d_{ijt} = 1|J^a_{ijt}, X_{ijt}; \beta) = \Phi(exp(\frac{1}{2}X'_i\beta)(\delta E(p_{j,t+1}|J_{ijt}) - (1 + r)p_{ijt}))$ as previously defined in equation 1. Obviously, to estimate β^*, the econometrician needs to compute $E(p_{ijt}|J^a_{ijt})$. To estimate β^*, the econometrician must specify the expectation $E(p_{j,t+1}|J^a_{ijt})$. Common approaches include assuming perfect foresight ($E(p_{j,t+1}|J^a_{ijt}) = p_{j,t+1}$) or rational expectations ($E(p_{j,t+1}|J^a_{ijt}) = F(J^a_{ijt})$). However, if the assumed information set is misspecified, the resulting parameter estimates will generally be biased.

Formally, the econometrician constructs a proxy for the VCs' expectations about the startup's future value: $E(p_{j,t+1}|J^a_{ijt}) = F(J^a_{ijt})$. When $J^a_{ijt} \neq J_{ijt}$, this leads to discrepancies between the VCs' true expectations and the econometrician's proxy: $E(p_{ijt}|J^a_{ijt}) = F(J^a_{ijt}) = F(J_{ijt}) + \epsilon = E(p_{ijt}|J_{ijt}) + \epsilon_{ijt}$. Whenever this error term ϵ_{ijt} is non-zero, the resulting estimates suffer from a bias akin to that in classical measurement error problems. A common solution is to instrument the problematic variable, which once again is constrained by the limited scope of VC-startup data.

5.2 Partial knowledge of information set

In most empirical settings, including the present one, econometricians cannot observe either agents' expectations or the exact information sets used in decision-making. For instance, while it is widely accepted that EBITDA is critical to firm valuation and thus likely to be part of a VC's information set, this variable is unfortunately missing from our data. Nevertheless, a subset of the information set is often observable. For example, it is reasonable to assume that VCs would expect higher growth from an AI startup during a boom in the AI sector. Sector-level indicators such as industry performance are likely observed by both VCs and researchers. While these variables do not capture the full information set, they plausibly influence expectations and can serve as partial proxies.

To address this issue, we adopt a moment inequality estimation procedure introduced by Dickstein and Morales (2018), which relies only on a subset of the information set used by agents to form expectations. Specifically, we assume $Z_{ijt} \subset J_{ijt}$. This estimation strategy offers several advantages. First, it enables identification of a confidence region for β^* that is robust to the biases inherent in misspecified maximum likelihood estimation. Second, it allows us to test which variables are likely included in the information sets of different VCs, potentially explaining heterogeneity in predictive accuracy. Finally, it facilitates the simulation and evaluation of counterfactual scenarios where broader or more accurate information sets are made available to agents. In what follows, we outline the key steps in the derivation of the moment inequalities.

5.2.1 Moment inequalities

The first group of inequalities is odds-based moment inequalities. Let $L(d_{ijt}|J_{ijt}, X; \beta)$ denote the log-likelihood conditional on J_{ijt}. Then the score function can be rearranged to be:

$$\begin{aligned}\frac{\partial L(d_{ijt}|J_{ijt}, X; \beta)}{\partial \beta} &= E(d_{ijt}\frac{1-\Phi(exp(\frac{1}{2}X_i'\beta)(\delta E(p_{j,t+1}|J_{ijt}) - (1+r)p_{ijt}))}{\Phi(exp(\frac{1}{2}X_i'\beta)(\delta E(p_{j,t+1}|J_{ijt}) - (1+r)p_{ijt}))} - (1-d_{ijt})|J_{ijt}, X) \\ &= 0.\end{aligned}$$

Under the assumption that the expectational error ν_{ijt} is mean zero and the property that $\frac{1-\Phi}{\Phi}$ is convex, one can apply Jensen's inequality to derive

$$E(d_{ijt}\frac{1-\Phi(exp(\frac{1}{2}X_i'\beta)(\delta p_{j,t+1}-(1+r)p_{ijt}))}{\Phi(exp(\frac{1}{2}X_i'\beta)(\delta p_{j,t+1}-(1+r)p_{ijt}))}-(1-d_{ijt})|J_{ijt},X)\geq 0.$$

By the Law of Iterated Expectations, replace the full information set J_{ijt} with the subset Z_{ijt}, we can get one set of the odds-based moment functions m_u^{ob}:

$$\begin{aligned} m_u^{ob} &= d_{ijt}\frac{1-\Phi(exp(\frac{1}{2}X_i'\beta)(\delta p_{j,t+1}-(1+r)p_{ijt}))}{\Phi(exp(\frac{1}{2}X_i'\beta)(\delta p_{j,t+1}-(1+r)p_{ijt}))}-(1-d_{ijt}),\\ m_l^{ob} &= (1-d_{ijt})\frac{\Phi(exp(\frac{1}{2}X_i'\beta)(\delta p_{j,t+1}-(1+r)p_{ijt}))}{1-\Phi(exp(\frac{1}{2}X_i'\beta)(\delta p_{j,t+1}-(1+r)p_{ijt}))}-d_{ijt}, \end{aligned} \tag{4}$$

where m_l^{ob} is derived in a similar pattern as m_u^{ob}. We define the conditional odds-based moment inequalities as

$$\begin{aligned} E(m_u^{ob}|Z_{ijt};\beta)&\geq 0,\\ E(m_l^{ob}|Z_{ijt};\beta)&\geq 0. \end{aligned}$$

The second set of moment inequalities derives from the revealed preference argument that a VC will invest if and only if the expected return surpasses the VC required return such that $E(d_{ijt} * (\delta E(p_{j,t+1}|J_{ijt}) + \nu_{ijt} - (1+r)p_{ijt})) \geq 0$. With similar algebra as the derivation of odds-based moments, we can get the revealed preference moment equations:

$$\begin{aligned} m_u^{ob} &= d_{ijt}(\delta p_{j,t+1}-(1+r)p_{ijt})+(1-d_{ijt})exp(-\frac{1}{2}X'\beta)\frac{\phi(exp(\frac{1}{2}X_i'\beta)(\delta p_{j,t+1}-(1+r)p_{ijt}))}{1-\Phi(exp(\frac{1}{2}X_i'\beta)(\delta p_{j,t+1}-(1+r)p_{ijt}))},\\ m_l^{ob} &= -(1-d_{ijt})(\delta p_{j,t+1}-(1+r)p_{ijt})+d_{ijt}exp(-\frac{1}{2}X'\beta)\frac{\phi(exp(\frac{1}{2}X_i'\beta)(\delta p_{j,t+1}-(1+r)p_{ijt}))}{\Phi(exp(\frac{1}{2}X_i'\beta)(\delta p_{j,t+1}-(1+r)p_{ijt}))}, \end{aligned} \tag{5}$$

and the corresponding moment equalities:

$$E(m_u^{ob}|Z_{ijt};\beta) \geq 0,$$
$$E(m_l^{ob}|Z_{ijt};\beta) \geq 0.$$

In the estimation procedure, we combine odds-based and revealed preference moment inequalities. The set defined by the odds-based inequalities is a singleton only under two restrictive conditions: when firms make no expectation errors (in the absence of the private signal ν_{ijt}), and when the vector of instruments Z_{ijt} fully captures the agents' information set. Under these conditions, the revealed preference inequalities offer no additional identifying power. However, in all other cases, the revealed preference inequalities contribute additional identification by leveraging observed choices that are consistent with utility-maximizing behavior under uncertainty.

5.2.2 Estimation implementation

The conditional moment inequalities cannot be directly used in empirical applications because conditioning on every possible value of Z_{ijt} is computationally infeasible. The standard approach, which we adopt, transforms these conditional moments into unconditional moment inequalities:

$$E\left[\begin{pmatrix} m_u^{ob} \\ m_l^{ob} \\ m_u^{rp} \\ m_l^{rp} \end{pmatrix} \times g(Z_{ijt})\right] \geq 0, \tag{6}$$

where the instrument function $g(Z_{ijt})$ consists of indicator functions (dummies) that partition the support of Z_{ijt} into quadrants.

To construct a confidence set for the true parameter β^*, we implement the asymptotic Generalized Moment Selection (GMS) procedure proposed by Andrews and Soares (2010). This procedure is based on the Modified Method of Moments (MMM) statistic. Specifically, let the set of inequalities be $\bar{m}_k(\beta) = \frac{1}{n}\sum_{ijt} m_k(X_{ijt}, Z_{ijt}, \beta) \geq 0, k = 1, 2, ..., K$. The MMM statistics is then defined

as

$$Q(\beta) = \sum_k (min(\frac{\bar{m}_k(\beta)}{\hat{\sigma}_k(\beta)}), 0)^2, \tag{7}$$

where $\hat{\sigma_k}(\beta) = \sqrt{\frac{1}{n}\sum_{ijt}(m_k(X_{ijt}, Z_{ijt}, \beta)) - \bar{m}_k(X_{ijt}, Z_{ijt}, \beta))^2}$. If $Q(\beta) = 0$, it indicates that all moment inequalities are satisfied at β. Otherwise, some sample moment inequalities are violated either because the true population moment inequalities fail to hold at β or due to sampling variation.

To construct the confidence set, we begin with a sufficiently large orthotope in the parameter space, with dimensionality equal to the number of scalar parameters in β. For each discretized point β_p in the grid, we test the null hypothesis that this vector equals the true value of β:

$$H_0 : \beta^* = \beta_p; \quad H_1 : \beta^* \neq \beta_p$$

by evaluating the MMM test statistic $Q(\beta_p)$ as defined in equation 7.

To determine the critical value, we simulate R draws[3] from multivariate normal distribution $N(0_K, I_K)$. Each draw ψ_r is used to compute the criterion function

$$Q^{AA}_{n,r} = \sum_k \{(min\{[\hat{\Omega}^{\frac{1}{2}}(\beta_p)\psi_r]_k, 0\}) \times \mathbf{1}\{\sqrt{n}\frac{\bar{m}_k(\beta_p)}{\hat{\sigma}_k(\beta_p)} \leq \sqrt{\ln n}\}\}$$

where $[\hat{\Omega}^{\frac{1}{2}}(\beta_p)\psi_r]_k$ denotes the kth element of the vector $\hat{\Omega}^{\frac{1}{2}}(\beta_p)\psi_r$, and $\hat{\Omega}^{\frac{1}{2}}(\beta_p)$ is the estiamted variance-covariance matrix of $\frac{m_k(\beta_p)}{\hat{\sigma_k}(\beta_p)}$.

The critical value $\hat{c}^{AA}_n(\beta_p, 1-\alpha)$ is the $(1-\alpha)$-quantile of the simulated distribution of $Q^{AA}_{n,r}(\beta_p)$ across the R draws. If the MMM statistic $Q(\beta_p) \leq \hat{c}^{AA}_n$, then we include β_p in the $(1-\alpha)\%$ confidence set. This testing procedure is repeated over all β_p in the orthotope grid. In our main estimation, we set $\alpha = 0.05$ and report the 95% confidence sets.

[3] We use R=1000.

6 Results

The baseline information set that we assume all VCs use to form expectations includes the startup's own price, the sector-year average price, and the round-year average price, denoted as $\{p_{ijt}, P_{\text{sector-year}}, P_{\text{t-year}}\}$. The rationale for this specification is as follows. The startup's own price reflects the present value of expected future profits and is thus a strong predictor of future price. The sector-year average price captures broader industry valuation trends, based on the assumption that startups within the same sector are subject to common shocks. The round-year average price accounts for lifecycle effects, recognizing that valuations may systematically differ across funding stages. Using this information set, we nonparametrically estimate the relationship between future prices and these variables, and interpret the predicted values as the systematic component of VCs' expectations.

Table 5 reports the estimation results. The first row presents maximum likelihood estimates under the unrealistic assumption of perfect foresight—that VCs know the future price with certainty. Although this assumption is clearly contradicted by the observed variation in investment performance, it serves as a useful benchmark to understand the bias of MLE estimates. Interestingly, all three VC characteristics exhibit negative coefficients under this specification, suggesting that more experience or a wider network is associated with worse predictive accuracy. Replacing perfect foresight with a price prediction proxy does not qualitatively change these results.

In contrast, the moment inequality approach yields a different pattern. Rows (3) and (4) of Table 5 present confidence intervals for the coefficients using both odds-based and revealed preference moments, and using only odds-based moments, respectively. Both approaches suggest that experience improves predictive accuracy and that corporate-backed VCs have weaker predictive performance. The full moment set also indicates that a wider network improves predictive accuracy, though the odds-only specification yields a confidence interval that is too wide to reject a null effect. From a theoretical standpoint, revealed preference moments provide no additional identification if the decision maker forms perfectly accurate expectations and uses the same instruments as those included in our estimation. The difference between the estimates in Rows (3) and (4) thus suggests that VCs use additional information beyond our baseline variables to infer startup

	CVC	Network	Experience
	(1)	(2)	(3)
MLE(PF)	-59.2	-0.1	-0.2
	(0.03)	(0.01)	(0.01)
MLE(E(P))	-64.9	0.00	-0.9
	(0.26)	(0.10)	(0.09)
MI(OB&RP)	(-15.6,-2.1)	(2.2,5.5)	(2.8,10.9)
MI(OB)	(-15.9,-0.1)	(-1.3,5.5)	(2.4,12.0)

Note: The figure reports the estimation results. The standard deviation of VC private signal is $exp(-\frac{1}{2}X'\beta)$. Column 1 represents corporate-backed VC indicator, column 2 represents the age of VC at the time of deal occurrence. Row 1 reports maximum likelihood estimate assuming perfect foresight; Row 2 reports maximum likelihood estimate assuming baseline information; Row 3 reports moment inequalities confidence sets using both odds-based and revealed preference conditions; Row 4 reports moment inequalities confidence sets using only odds-based conditions.

Table 5: Estimation results

value—a realistic and expected feature of the data.

7 Conclusion

In this paper, we investigate how the predictive power of venture capitalists varies with their characteristics. Using data on VC investment history, we first present reduced-form evidence on the correlation between VC characteristics and their investment performance.

We then develop a behavioral model of the venture capital market. A key assumption in the model is the absence of strategic interactions among VCs and the neglect of winner's curse considerations. While not definitive, we provide suggestive evidence that these simplifications are not overly restrictive for our analysis.

Our structural estimation focuses on how VC characteristics influence predictive accuracy, which we measure through the variation in private signals. We find that predictive accuracy improves with greater experience and increased collaboration among VCs. These results suggest that policies encouraging VC investment may generate a virtuous cycle, with larger welfare gains over the long run.

Finally, our empirical findings imply that the set of instruments we employ differs from the information VCs actually use to form expectations. Future research could investigate the nature of this information set and help answer the question: What do VCs know?

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